orthonormal bases suitable for computing the overlap metric of Section H.2.2.

Why mean-centering works. Because the factors are independent, activations within each fixed configuration differ only due to variation in factor n. Mean-centering removes the contribution of the fixed factors, isolating the geometry associated with factor n.

Checking factoredness. The effective dimensionality of each factor's vary-one activations also provides a check on the degree of factoredness: for a fully factored representation, the effective dimensionality should match the intrinsic dimensionality of that factor's ground-truth belief simplex ($d_n - 1$). Higher effective dimensionality suggests the representation is not fully factored.

Limitations. Vary-one analysis requires the ability to generate samples by fixing the latent dynamics of all but one factor—something that is straightforward for independent factors but not for processes with conditional dependencies among factors. For such settings, we turn to regression-based subspace identification.

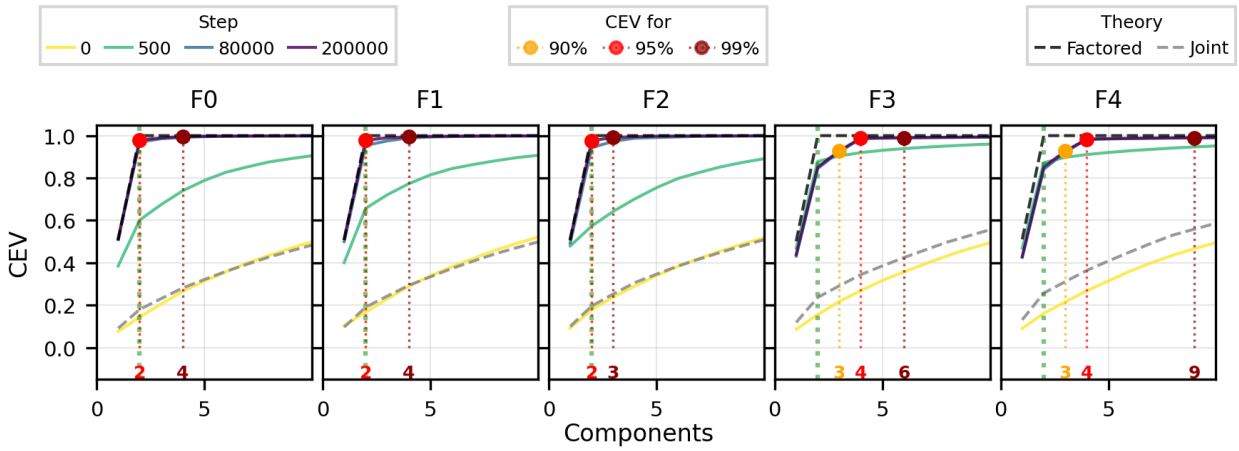

Figure 15. The CEV curves for residual stream activations when driven by the vary-one dataset for each factor. The vertical dashed green line represents the predicted dimensionality (2 for each factor, in this case). **As predicted, the first two components are responsible for much more variation than the rest for all factors**. Factors 3 and 4 ultimately spread out their activations across more dimensions, but still reach a CEV of 95% with only 4 PCs.

H.1.2. Regression-Based Subspace Identification

While vary-one analysis provides a direct method for identifying factor subspaces through controlled variation, it requires the ability to generate samples by fixing the latent dynamics of all but one factor in the generative process—something that is straightforward for generative processes with independent factors but not for those with conditional dependencies among factors. In such settings, we can instead leverage the regression framework of Section G to identify factor subspaces.

As described in Section G.4, we perform a single joint regression from activations to the concatenated predictive vector $Y = [Y^{(1)}|\cdots|Y^{(N)}]$, yielding coefficient matrix $W = [W^{(1)}|\cdots|W^{(N)}]$ where $W^{(n)} \in \mathbb{R}^{d_{\text{model}} \times d_n}$ corresponds to factor n. We define the **factor subspace** $\mathcal{U}^{(n)}$ as the subspace spanned by the columns of $W^{(n)}$. To obtain an orthonormal basis for $\mathcal{U}^{(n)}$, we compute the SVD of $W^{(n)}$ and retain the left singular vectors corresponding to singular values above a numerical tolerance, yielding $Q^{(n)} \in \mathbb{R}^{d_{\text{model}} \times r_n}$ where $r_n \leq d_n$ is the effective rank. Orthonormalization is necessary because the overlap metric in Section H.2.2 relies on principal angles between subspaces, which are only well-defined with respect to orthonormal bases; without this step, the metric would conflate geometric alignment with the arbitrary scaling and skewness of the regression coefficients.

Intuitively, $\mathcal{U}^{(n)}$ captures the directions in activation space that carry information predictive of factor n's predictive vector. If the transformer has learned a factored representation, we expect $\mathcal{U}^{(n)}$ to be approximately orthogonal to $\mathcal{U}^{(m)}$ for $n \neq m$. Note, however, that successful linear regression—low reconstruction error—does not by itself imply orthogonality; a representation in which factor subspaces overlap could still support accurate linear readout of each factor's predictive vector.

H.2. Orthogonality Analysis

H.2.1. Effective Dimensionality Comparison

We offer a global perspective on subspace orthogonality that leverages the relationship between the dimensionality of factor-specific representations and the dimensionality of their combined representation. For subspaces $\mathcal{A}$ and $\mathcal{B}$ of a common vector space, a standard result from linear algebra gives:

$$\dim(\mathcal{A}+\mathcal{B}) = \dim(\mathcal{A}) + \dim(\mathcal{B}) - \dim(\mathcal{A}\cap\mathcal{B}) \tag{42}$$

where $\mathcal{A}+\mathcal{B}$ denotes the subspace spanned by the union of $\mathcal{A}$ and $\mathcal{B}$. Thus, dimensionalities of the subspaces are additive only when $\mathcal{A}\cap\mathcal{B} = \{0\}$, i.e., when the subspaces are orthogonal. Rearranging Eq. (42), we see that $\dim(\mathcal{A}) + \dim(\mathcal{B}) - \dim(\mathcal{A}+\mathcal{B}) = \dim(\mathcal{A}\cap\mathcal{B})$, so any deviation from additivity directly measures the dimension of overlap of these subspaces.

In practice, activations form point clouds that approximately span low-dimensional subspaces rather than lying exactly on them. We therefore use the effective dimensionality k_p^* as an empirical proxy for subspace dimension. For a transformer trained on a generative process with N factors, we compute:

1. $k_p^{*(n)}$: the effective dimensionality of activations generated from vary-one datasets for factor n, mean-centered per frozen context as described in Section H.1.1
2. $k_p^{*(\text{all})}$: the effective dimensionality of activations from the union of all vary-one datasets.

If the factor subspaces are orthogonal, the sum $\sum_{n=1}^{N} k_p^{*(n)}$ should equal $k_p^{*(\text{all})}$. If the sum exceeds $k_p^{*(\text{all})}$, the subspaces must share directions. The magnitude of this discrepancy provides a global measure of overlap that complements the pairwise subspace overlap metric of Section H.2.2. While the overlap metric examines geometric relationships between explicitly identified bases, the dimensionality comparison requires only the effective dimensionality of activation point clouds, avoiding the need to commit to specific basis vectors.

The choice of CEV threshold p introduces some arbitrariness into this analysis. In practice, we report results using $p = 0.95$ as we find that qualitative conclusions are robust across reasonable threshold choices (e.g., $p \in [0.90, 0.99]$) and that $p = 0.95$ more than captures the bulk of the variance associated with a given factor's predictive geometry.

H.2.2. Subspace Overlap Metric

We provide a second, pairwise perspective by quantifying the orthogonality between two candidate subspaces $\mathcal{A}$ and $\mathcal{B}$ via their principal angles (also known as canonical angles). Given orthonormal bases $Q_{\mathcal{A}}$ and $Q_{\mathcal{B}}$ for each subspace (obtained via the procedures in Section H.1), we compute the singular value decomposition of their interaction matrix $M = Q_{\mathcal{A}}^\top Q_{\mathcal{B}}$. The singular values σ_i of M correspond to the cosines of the *principal angles* θ_i between $\mathcal{A}$ and $\mathcal{B}$. We define the **subspace overlap** as:

$$\text{overlap}(\mathcal{A},\mathcal{B}) = \frac{1}{d_{\min}}\sum_{i=1}^{d_{\min}} \sigma_i^2 = \frac{1}{d_{\min}}\|M\|_{\text{F}}^2 = \frac{1}{d_{\min}}\text{Tr}(P_{\mathcal{A}}P_{\mathcal{B}}) \tag{43}$$

where $d_{\min} = \min(d_{\mathcal{A}}, d_{\mathcal{B}})$ is the dimension of the smaller subspace and $P_{\mathcal{Y}} = Q_{\mathcal{Y}}Q_{\mathcal{Y}}^\top$ is the orthogonal projection onto subspace $\mathcal{Y}$. This yields an overlap score between 0 and 1, which is a basis-invariant measure of the shared dimensionality between the subspaces, where 1 indicates that the smaller subspace is fully contained within the larger one, and 0 indicates that the subspaces are geometrically orthogonal. As suggested in Eq. (43), there are many ways to interpret this metric and it is mathematically equivalent to 1) the squared Frobenius norm of M scaled by $\frac{1}{d_{\min}}$ and 2) the normalized trace of the product of the projection matrices, $\frac{1}{d_{\min}}\text{Tr}(P_{\mathcal{A}}P_{\mathcal{B}})$.

I. Empirical synopsis

We find that neural networks do indeed learn to factor processes into parts, with latent distributions over these parts updated in parallel. We found (from CEV analysis) that only the low number of dimensions are used by these trained models. And we found that a linear map exists from these activations to the anticipated belief geometries in orthogonal subspaces, as

suggested by the RMSE (reducing over the course of training together with loss) between the ground-truth geometry of the factors and the linear regression of activations to this geometry. Even without linear regression, the anticipated geometry can be seen in the relevant 2-dimensional subspaces defined by pairs of principal vectors.